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Sleep Stage Monitoring

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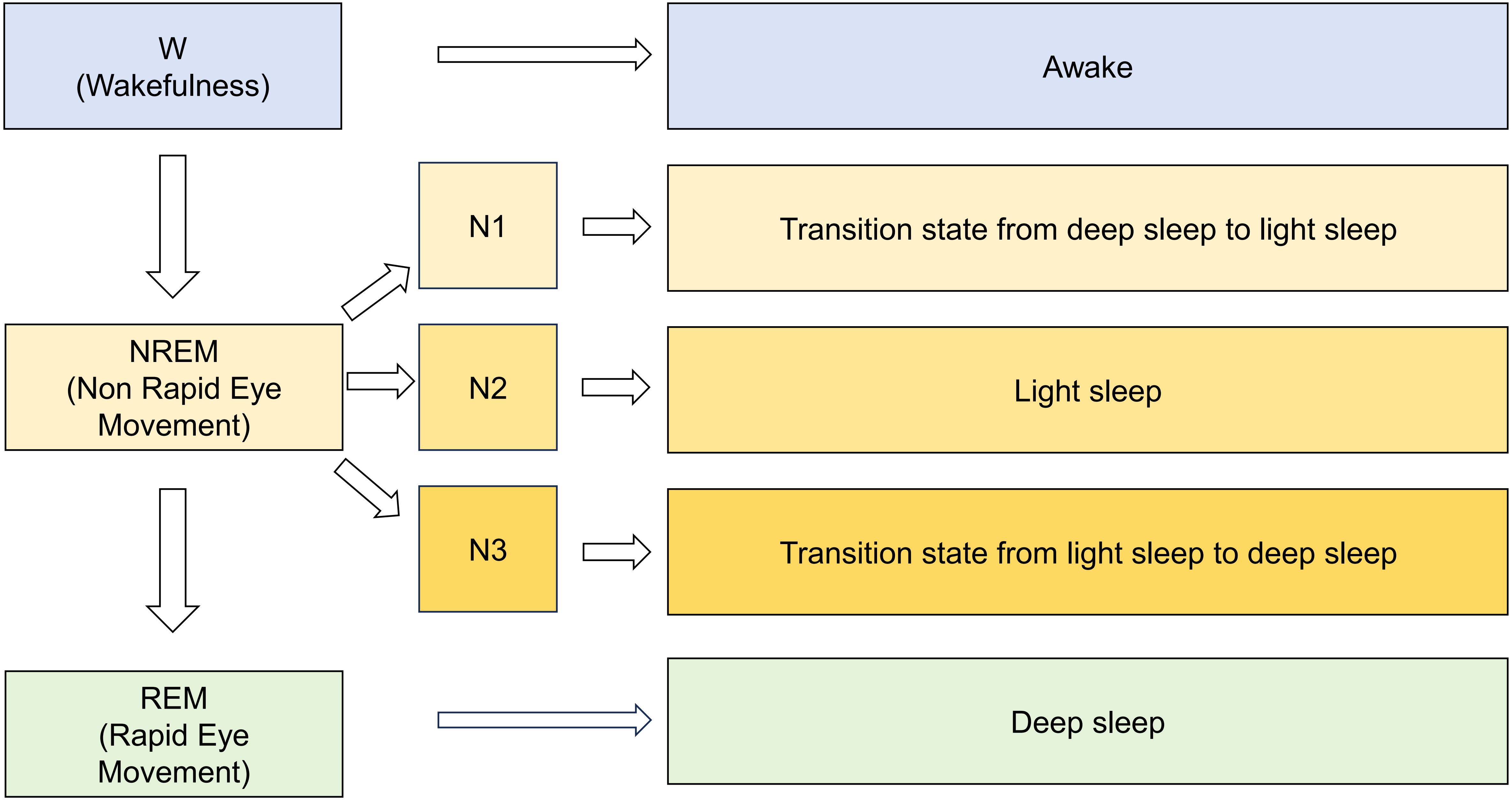
Problem Statement

- **Poor sleep quality** and irregular sleep patterns can **negatively affect** daily productivity, concentration, and overall well-being.
- **Conventional alarms** wake users at a fixed time, which may result in waking during deeper sleep leading to **grogginess**.
- **Raw physiological** and **movement sensor data** is difficult to interpret without intelligent processing.
- Existing sleep-monitoring solutions often provide sleep data or reports but **do not provide a personalized wake-up decision** based on the user's current sleep stage and preferred wake-up window.

Need for Solution

- A **low-cost, accessible** and intelligent sleep-monitoring system.
- Convert **raw sensor readings** into meaningful sleep information.
- Provide a **smart wake-up mechanism** within the user's preferred wake-up window.

The 4 Sleep Stages



Drawbacks of 4 Sleep Stages

- **Higher model complexity** - The model has to distinguish among N1, N2, N3, and REM instead of making a simpler target decision.
- **Requires more representative training data** - The model needs enough labeled examples of all four stages across different people and sleep conditions.
- **Greater sensitivity to individual differences** - Heart rate, HRV, movement, and EEG patterns can vary considerably between people, making a four-class model harder to generalize.
- **Stage transitions are not always clear-cut** - Sleep stages change gradually, so an epoch around a transition may contain characteristics of more than one stage.

References

https://link.springer.com/article/10.1007/s10462-024-10926-9

Automatic sleep stage classification using deep learning: signals, data representation, and neural networks

Our Solution to this Drawback

4 Sleep Stages
(N1, N2, N3, REM)

Signal Requirements
PSG -> EEG + EOG + EMG + Physiological signals

Our Low Cost Sensor Setup
MAX30102 → PPG → HR, HRV, SpO_2 MPU6050 → Movement

Narrowed ML Objective
N2 vs Non- N2

Smart Wake
Detect sustained N2 within the user's wake window

References

https://royalsocietypublishing.org/rsos/article/10/4/221517/91905/
Multi-stage sleep classification using photoplethysmographic sensor
https://arxiv.org/pdf/2305.02731
A Cluster-Based Opposition Differential Evolution Algorithm Boosted by a Local Search for ECG Signal Classification

Algorithm and Formula

MAX30102 + MPU6050
Sensor hardware

→ **Continuous sensor data**
HR, SpO2, movement

→ **30-second epoch**
Groups raw readings

→ **Feature calculation**
HR, HRV, SpO2, MOV

→ **ML N2 classifier**
Trained model

→ **N2 probability**
Score 0 to 1

→ **Threshold check**
N2 or not N2

→ **Consecutive epoch check**
Filters noise

→ **Wake window check**
Inside user range?

→ **WAKE / CONTINUE**
Or force wake

Sensors

MAX30102: heart rate, RR intervals (for HRV), SpO2
MPU6050: body movement

What is an epoch?

One epoch = 30 seconds of continuous sensor data. An epoch is a time window created by grouping many raw sensor readings — the sensor does not directly produce an "epoch."

How One Epoch becomes an ML Input?

30-second raw sensor data → **Calculate features** → **Feature vector**

HR

Average heart rate during the epoch.

HR = 61.4 BPM

HRV

Heart rate variability from beat-to-beat RR intervals (RMSSD).

$$HRVi = \sqrt{\left[\Sigma(RR_{n+1} - RR_n)^2 / (N-1) \right]}$$

HRV = 42 ms

SpO2

Average blood oxygen saturation during the epoch.

SpO2 = 98%

Movement

Derived from MPU6050 acceleration. Low movement → small value.

MOV = 0.08

Feature vector: `xi = [ HRi, HRVi, SpO2,i, MOVi ]` **Example:** `x1 = [61.4, 42, 98, 0.08]`

Feature = one numerical measurement. **Feature vector** = all the features of ONE epoch grouped together.

Epoch	HR	HRV	SpO2	Movement
1	61.4	42	98	0.08
2	60.8	39	98	0.05

N2 Prediction and Stability Check

Feature vector → Trained ML model → N2 probability pi

$$p_i = f_\varphi(x_i)$$

The trained model outputs a score from 0 to 1: how strongly this epoch resembles N2.

Example: $x_1 = [61.4,\ 42,\ 98,\ 0.08] \rightarrow p_1 = 0.81$

Threshold check — θ = 0.70 (initial example value)

$d_i = 1$ if $p_i \geq \theta$, otherwise 0

Epoch	N2 probability	di
1	0.62	0
2	0.81	1
3	0.75	1
4	0.88	1

θ = 0.70 is NOT scientifically fixed — it must be tuned using validation data.

Stability check — k = 3 consecutive N2 epochs

$S_i = \Sigma\, d_j$ over the last k epochs. If $S_i = k \rightarrow$ N2 CONFIRMED

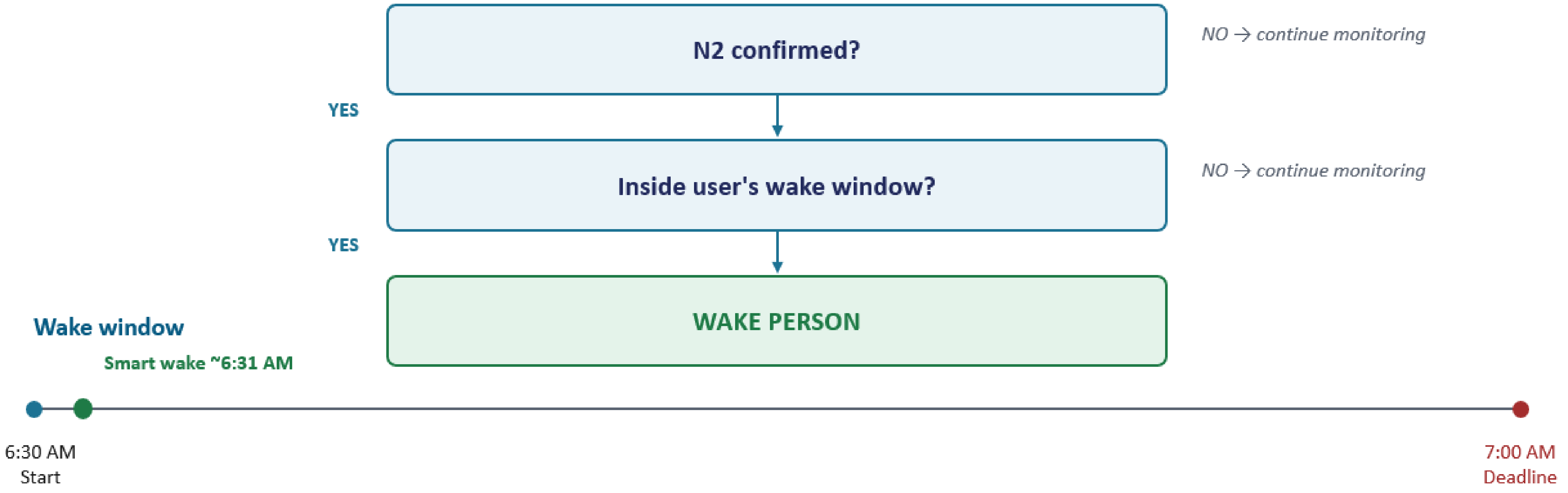
Worked example: $d_2 + d_3 + d_4 = 1 + 1 + 1 = 3$
$k = 3 \rightarrow$ **N2 CONFIRMED**

“Consecutive” means immediately adjacent epochs are all N2 — not just any three N2 epochs scattered across the night.

N2 → N2 → N2 ✓ CONFIRMED

N2 → NOT N2 → N2 ✗ NOT CONFIRMED

Final Wake Decision



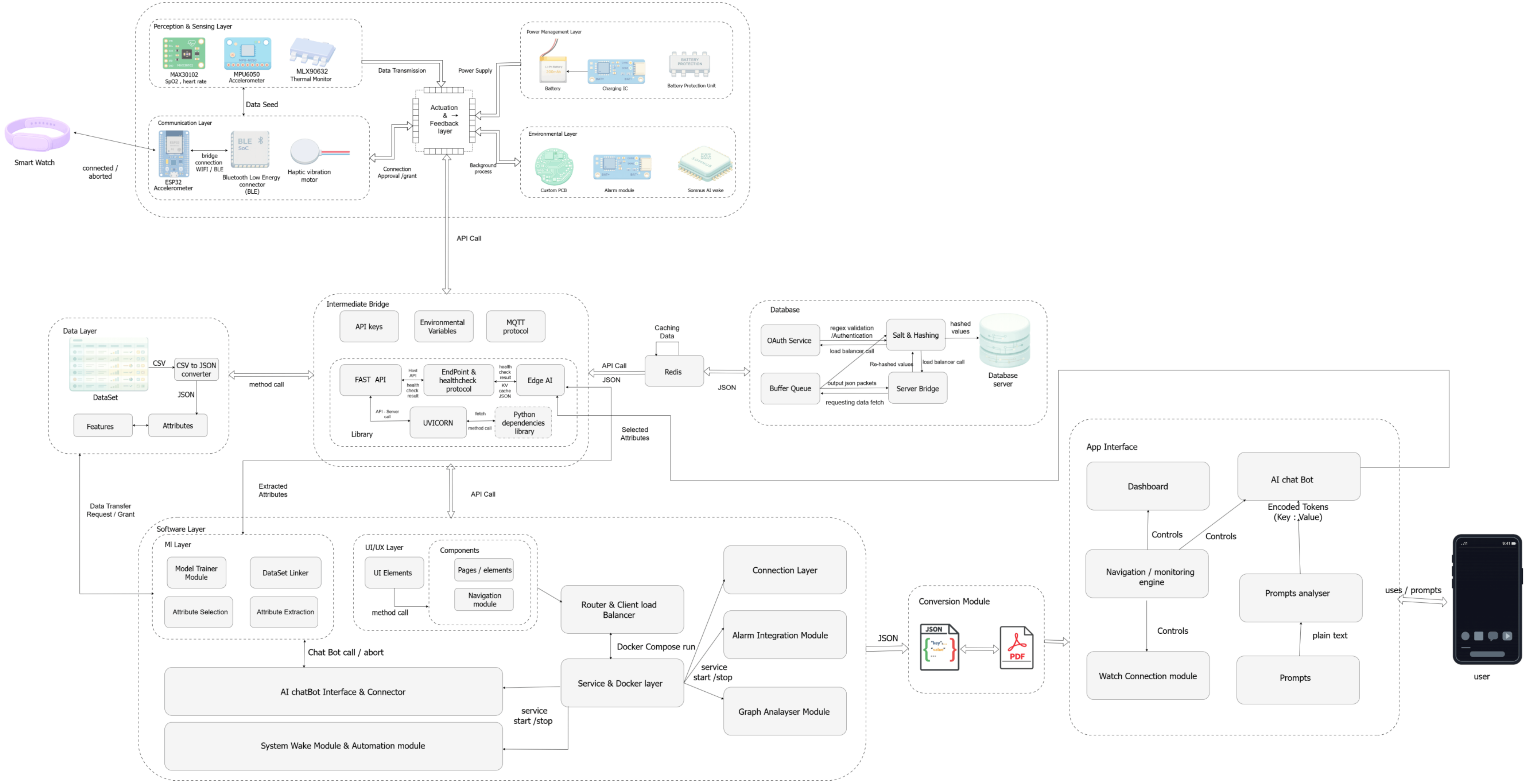

```
Wakei = 1   if Confirmedi = 1  AND  Tstart ≤ ti ≤ Tdeadline
Wakei = 1   if ti = Tdeadline   (forced fallback)
Wakei = 0   otherwise
```

Example (epoch = 30s, θ = 0.70, k = 3, window 6:30–7:00): `6:30→0.81→N2, 6:30:30→0.76→N2, 6:31→0.83→N2 → 3×30s = 90s →` **N2 CONFIRMED, WAKE ~6:31 AM**

If no N2 is confirmed by 7:00 AM → FORCE WAKE at 7:00 AM regardless of sleep stage.

Raw sensor data → 30-s epoch → features → ML N2 probability → threshold → consecutive N2 → smart wake / deadline fallback

Architecture Diagram



Prototype Sensors and Cost

Component	Purpose	Approx. Cost / Unit
MAX30102 breakout module	PPG, HR, HRV, SpO_2	₹380 – ₹765
MPU6050 breakout module	Accelerometer and gyroscope	₹290 – ₹670
ESP32 Dev Board	Main controller and Wi-Fi/BLE	₹480 – ₹960
LiPo Battery and TP4056	Power and charging	₹380 – ₹575
Carrier PCB	Mounting or interconnection	₹190 – ₹960
Wires, connectors, strap and enclosure	Assembly and wearable packaging	₹480 – ₹960

Estimated Total / Prototype ≈ **₹2,400 – ₹4,900**

Custom PCBA for the Prototype Sensors

Component	Purpose	Approx. Cost / Unit
PCB Design & Layout	Schematic, PCB layout, routing	₹5,000 – ₹12,000
PCB Fabrication	Custom fabricated boards	₹1,500 – ₹4,000 / board
PCBA Assembly	SMT/component placement or soldering	₹1,500 – ₹4,000 / board
MAX30102 + MPU6050	Sensor ICs/modules/components	₹500 – ₹1,200 / board
ESP32	ESP32 module or supporting circuitry	₹500 – ₹1,000 / board
Power Circuit	LiPo charging and regulation	₹300 – ₹800 / board
Connectors	Battery connector, headers, etc.	₹300 – ₹700 / board
Enclosure / Strap	Wearable packaging	₹500 – ₹1,500 / unit

Estimated Total / Prototype ≈ **₹ 30,000 – ₹60,000**

Our Progress

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

PS C:\Users\f4own\Projects\sleepsense_ai\dataset\src> python train_baseline.py
CLASSIFICATION REPORT
======================================
              precision    recall  f1-score   support

          N1     0.1030    0.1653    0.1269      1543
          N2     0.6831    0.5761    0.6250      8559
          N3     0.3103    0.0247    0.0458       364
           R     0.1514    0.1729    0.1615      1822
           W     0.3606    0.4302    0.3923      2513

    accuracy                         0.4453     14801
   macro avg     0.3217    0.2738    0.2703     14801
weighted avg     0.4932    0.4453    0.4623     14801
```

```
======================================
CONFUSION MATRIX
======================================
      N1    N2   N3     R     W
N1   255   498    0   276   514
N2  1225  4931   19  1193  1191
N3    39   216    9    63    37
R    479   853    0   315   175
W    477   721    1   233  1081
```

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

PS C:\Users\f4own\Projects\sleepsense_ai\dataset\src> python train_xgboost.py
======================================
TOP 30 FEATURE IMPORTANCE
======================================
                 Feature  Importance
                HR_range    0.039685
           HR_range_prev    0.036845
  Movement_std_delta_prev   0.032782
               ACC_Y_std    0.030115
               ACC_X_std    0.027799
           HR_range_next    0.019577
  Movement_rms_delta_prev   0.017694
              ACC_Y_mean    0.017298
  Movement_rms_delta_next   0.017215
Movement_energy_delta_prev  0.016482
```

```
PS C:\Users\f4own\Projects\sleepsense_ai\dataset\src> python train_xgboost.py
            TEMP_mean_prev    0.012354
                    HR_max    0.012298
                  TEMP_min    0.011864
                   EDA_min    0.011744
  Movement_mean_delta_prev    0.011523
         Movement_rms_prev    0.011449
         Movement_std_prev    0.011321
Movement_energy_delta_next    0.011095
           Movement_energy    0.011038
      TEMP_mean_delta_next    0.010856
                  IBI_mean    0.010762
             IBI_mean_next    0.010746
                   EDA_max    0.010725
PS C:\Users\f4own\Projects\sleepsense_ai\dataset\src>
```

Thank You